

Primary 1-3 TEACHER PRACTICAL ACTIVITY MANUAL

Implementation guide for PA-101 to PA-200

OBSERVE -> MEASURE -> COMPARE -> GROW -> RECORD

Purpose	Turn the GrowMeal practical cards into safe, repeatable, age-appropriate teaching sessions.
Teacher role	Prepare the site, demonstrate technique, supervise safety, question learners, check evidence and connect the activity to the journal, mission cards and assessment system.
Core cycle	PREPARE -> DEMONSTRATE -> PRACTISE -> RECORD -> DISCUSS -> CONNECT -> CLEAN UP
Assessment	Use observation, student records and the linked practical/investigation evidence. Avoid grading recall alone.
Local adaptation	Substitute locally available materials when the learning objective, safety and measurement quality are preserved.

WHOLE-MANUAL SAFETY & IMPLEMENTATION STANDARD

Teachers must inspect the garden before each activity; establish boundaries; match tools to learner age; supervise water, cutting tools, electrical equipment, pumps, nutrient solutions and instruments; require handwashing after garden work; prohibit tasting unless a separate approved food-safety activity explicitly permits it; stop any activity when conditions are unsafe.

Before class	Check site, weather/working conditions, tools, labels, water, access routes and learner grouping.
During class	Demonstrate first; keep learners within assigned zones; monitor tool use and protect plants, structures and instruments.
After class	Count/clean/store tools; close water systems; restore labels; record damage, abnormal readings or incidents; wash hands.
Research integrity	Never invent, overwrite or silently delete observations. Flag questionable data and document repeat/reference checks.
Accessibility	Use peer roles, large-print/visual instructions, reachable observation stations and alternative recording methods where appropriate.

GARDEN 01 - Alphabet Food Garden

Teaching focus: food-crop names and garden vocabulary.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Alphabet Food Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-101 - OBSERVE - Alphabet Food Garden

Learning objective	Enable pupils to observe food-crop names and garden vocabulary using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Alphabet Food Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils observe at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-102 - COUNT - Alphabet Food Garden

Learning objective	Enable pupils to count food-crop names and garden vocabulary using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Alphabet Food Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils count at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-103 - MEASURE - Alphabet Food Garden

Learning objective	Enable pupils to measure food-crop names and garden vocabulary using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Alphabet Food Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils measure at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-104 - COMPARE - Alphabet Food Garden

Learning objective	Enable pupils to compare food-crop names and garden vocabulary using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Alphabet Food Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils compare at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-105 - RECORD - Alphabet Food Garden

Learning objective	Enable pupils to record food-crop names and garden vocabulary using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Alphabet Food Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils record at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-106 - MAP - Alphabet Food Garden

Learning objective	Enable pupils to map food-crop names and garden vocabulary using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Alphabet Food Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils map at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-107 - GROW - Alphabet Food Garden

Learning objective	Enable pupils to grow food-crop names and garden vocabulary using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Alphabet Food Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils grow at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-108 - CHECK - Alphabet Food Garden

Learning objective	Enable pupils to check food-crop names and garden vocabulary using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Alphabet Food Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils check at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-109 - EXPLAIN - Alphabet Food Garden

Learning objective	Enable pupils to explain food-crop names and garden vocabulary using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Alphabet Food Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils explain at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-110 - IMPROVE - Alphabet Food Garden

Learning objective	Enable pupils to improve food-crop names and garden vocabulary using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Alphabet Food Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils improve at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 02 - Number Food Garden

Teaching focus: counts, plant height and simple totals.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Number Food Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-111 - OBSERVE - Number Food Garden

Learning objective	Enable pupils to observe counts, plant height and simple totals using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Number Food Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils observe at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-112 - COUNT - Number Food Garden

Learning objective	Enable pupils to count counts, plant height and simple totals using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Number Food Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils count at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-113 - MEASURE - Number Food Garden

Learning objective	Enable pupils to measure counts, plant height and simple totals using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Number Food Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils measure at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-114 - COMPARE - Number Food Garden

Learning objective	Enable pupils to compare counts, plant height and simple totals using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Number Food Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils compare at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-115 - RECORD - Number Food Garden

Learning objective	Enable pupils to record counts, plant height and simple totals using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Number Food Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils record at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-116 - MAP - Number Food Garden

Learning objective	Enable pupils to map counts, plant height and simple totals using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Number Food Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils map at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-117 - GROW - Number Food Garden

Learning objective	Enable pupils to grow counts, plant height and simple totals using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Number Food Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils grow at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-118 - CHECK - Number Food Garden

Learning objective	Enable pupils to check counts, plant height and simple totals using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Number Food Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils check at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-119 - EXPLAIN - Number Food Garden

Learning objective	Enable pupils to explain counts, plant height and simple totals using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Number Food Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils explain at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-120 - IMPROVE - Number Food Garden

Learning objective	Enable pupils to improve counts, plant height and simple totals using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Number Food Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils improve at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 03 - Geometry Garden

Teaching focus: shapes, sides, corners, length and garden layout.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Geometry Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-121 - OBSERVE - Geometry Garden

Learning objective	Enable pupils to observe shapes, sides, corners, length and garden layout using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Geometry Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils observe at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-122 - COUNT - Geometry Garden

Learning objective	Enable pupils to count shapes, sides, corners, length and garden layout using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Geometry Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils count at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-123 - MEASURE - Geometry Garden

Learning objective	Enable pupils to measure shapes, sides, corners, length and garden layout using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Geometry Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils measure at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-124 - COMPARE - Geometry Garden

Learning objective	Enable pupils to compare shapes, sides, corners, length and garden layout using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Geometry Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils compare at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-125 - RECORD - Geometry Garden

Learning objective	Enable pupils to record shapes, sides, corners, length and garden layout using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Geometry Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils record at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-126 - MAP - Geometry Garden

Learning objective	Enable pupils to map shapes, sides, corners, length and garden layout using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Geometry Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils map at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-127 - GROW - Geometry Garden

Learning objective	Enable pupils to grow shapes, sides, corners, length and garden layout using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Geometry Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils grow at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-128 - CHECK - Geometry Garden

Learning objective	Enable pupils to check shapes, sides, corners, length and garden layout using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Geometry Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils check at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-129 - EXPLAIN - Geometry Garden

Learning objective	Enable pupils to explain shapes, sides, corners, length and garden layout using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Geometry Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils explain at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-130 - IMPROVE - Geometry Garden

Learning objective	Enable pupils to improve shapes, sides, corners, length and garden layout using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Geometry Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils improve at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 04 - Leaf Garden

Teaching focus: leaf color, shape, size and texture.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Leaf Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-131 - OBSERVE - Leaf Garden

Learning objective	Enable pupils to observe leaf color, shape, size and texture using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Leaf Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils observe at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-132 - COUNT - Leaf Garden

Learning objective	Enable pupils to count leaf color, shape, size and texture using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Leaf Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils count at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-133 - MEASURE - Leaf Garden

Learning objective	Enable pupils to measure leaf color, shape, size and texture using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Leaf Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils measure at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-134 - COMPARE - Leaf Garden

Learning objective	Enable pupils to compare leaf color, shape, size and texture using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Leaf Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils compare at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-135 - RECORD - Leaf Garden

Learning objective	Enable pupils to record leaf color, shape, size and texture using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Leaf Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils record at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-136 - MAP - Leaf Garden

Learning objective	Enable pupils to map leaf color, shape, size and texture using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Leaf Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils map at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-137 - GROW - Leaf Garden

Learning objective	Enable pupils to grow leaf color, shape, size and texture using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Leaf Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils grow at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-138 - CHECK - Leaf Garden

Learning objective	Enable pupils to check leaf color, shape, size and texture using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Leaf Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils check at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-139 - EXPLAIN - Leaf Garden

Learning objective	Enable pupils to explain leaf color, shape, size and texture using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Leaf Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils explain at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-140 - IMPROVE - Leaf Garden

Learning objective	Enable pupils to improve leaf color, shape, size and texture using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Leaf Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils improve at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 05 - Pollinator Garden

Teaching focus: flowers and pollinator observations.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Pollinator Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-141 - OBSERVE - Pollinator Garden

Learning objective	Enable pupils to observe flowers and pollinator observations using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Pollinator Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils observe at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-142 - COUNT - Pollinator Garden

Learning objective	Enable pupils to count flowers and pollinator observations using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Pollinator Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils count at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-143 - MEASURE - Pollinator Garden

Learning objective	Enable pupils to measure flowers and pollinator observations using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Pollinator Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils measure at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-144 - COMPARE - Pollinator Garden

Learning objective	Enable pupils to compare flowers and pollinator observations using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Pollinator Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils compare at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-145 - RECORD - Pollinator Garden

Learning objective	Enable pupils to record flowers and pollinator observations using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Pollinator Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils record at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-146 - MAP - Pollinator Garden

Learning objective	Enable pupils to map flowers and pollinator observations using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Pollinator Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils map at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-147 - GROW - Pollinator Garden

Learning objective	Enable pupils to grow flowers and pollinator observations using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Pollinator Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils grow at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-148 - CHECK - Pollinator Garden

Learning objective	Enable pupils to check flowers and pollinator observations using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Pollinator Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils check at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-149 - EXPLAIN - Pollinator Garden

Learning objective	Enable pupils to explain flowers and pollinator observations using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Pollinator Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils explain at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-150 - IMPROVE - Pollinator Garden

Learning objective	Enable pupils to improve flowers and pollinator observations using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Pollinator Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils improve at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 06 - Raised Food Garden

Teaching focus: raised-bed growth, spacing and care.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Raised Food Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-151 - OBSERVE - Raised Food Garden

Learning objective	Enable pupils to observe raised-bed growth, spacing and care using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Raised Food Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils observe at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-152 - COUNT - Raised Food Garden

Learning objective	Enable pupils to count raised-bed growth, spacing and care using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Raised Food Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils count at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-153 - MEASURE - Raised Food Garden

Learning objective	Enable pupils to measure raised-bed growth, spacing and care using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Raised Food Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils measure at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-154 - COMPARE - Raised Food Garden

Learning objective	Enable pupils to compare raised-bed growth, spacing and care using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Raised Food Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils compare at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-155 - RECORD - Raised Food Garden

Learning objective	Enable pupils to record raised-bed growth, spacing and care using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Raised Food Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils record at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-156 - MAP - Raised Food Garden

Learning objective	Enable pupils to map raised-bed growth, spacing and care using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Raised Food Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils map at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-157 - GROW - Raised Food Garden

Learning objective	Enable pupils to grow raised-bed growth, spacing and care using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Raised Food Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils grow at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-158 - CHECK - Raised Food Garden

Learning objective	Enable pupils to check raised-bed growth, spacing and care using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Raised Food Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils check at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-159 - EXPLAIN - Raised Food Garden

Learning objective	Enable pupils to explain raised-bed growth, spacing and care using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Raised Food Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils explain at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-160 - IMPROVE - Raised Food Garden

Learning objective	Enable pupils to improve raised-bed growth, spacing and care using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Raised Food Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils improve at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 07 - ReBottle Wall Garden

Teaching focus: vertical container position, drainage and growth.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the ReBottle Wall Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-161 - OBSERVE - ReBottle Wall Garden

Learning objective	Enable pupils to observe vertical container position, drainage and growth using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the ReBottle Wall Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils observe at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-162 - COUNT - ReBottle Wall Garden

Learning objective	Enable pupils to count vertical container position, drainage and growth using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the ReBottle Wall Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils count at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-163 - MEASURE - ReBottle Wall Garden

Learning objective	Enable pupils to measure vertical container position, drainage and growth using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the ReBottle Wall Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils measure at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-164 - COMPARE - ReBottle Wall Garden

Learning objective	Enable pupils to compare vertical container position, drainage and growth using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the ReBottle Wall Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils compare at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-165 - RECORD - ReBottle Wall Garden

Learning objective	Enable pupils to record vertical container position, drainage and growth using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the ReBottle Wall Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils record at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-166 - MAP - ReBottle Wall Garden

Learning objective	Enable pupils to map vertical container position, drainage and growth using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the ReBottle Wall Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils map at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-167 - GROW - ReBottle Wall Garden

Learning objective	Enable pupils to grow vertical container position, drainage and growth using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the ReBottle Wall Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils grow at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-168 - CHECK - ReBottle Wall Garden

Learning objective	Enable pupils to check vertical container position, drainage and growth using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the ReBottle Wall Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils check at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-169 - EXPLAIN - ReBottle Wall Garden

Learning objective	Enable pupils to explain vertical container position, drainage and growth using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the ReBottle Wall Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils explain at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-170 - IMPROVE - ReBottle Wall Garden

Learning objective	Enable pupils to improve vertical container position, drainage and growth using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the ReBottle Wall Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils improve at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 08 - Sack GrowBag Garden

Teaching focus: growbag plant position, growth and care.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Sack GrowBag Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-171 - OBSERVE - Sack GrowBag Garden

Learning objective	Enable pupils to observe growbag plant position, growth and care using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Sack GrowBag Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils observe at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-172 - COUNT - Sack GrowBag Garden

Learning objective	Enable pupils to count growbag plant position, growth and care using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Sack GrowBag Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils count at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-173 - MEASURE - Sack GrowBag Garden

Learning objective	Enable pupils to measure growbag plant position, growth and care using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Sack GrowBag Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils measure at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-174 - COMPARE - Sack GrowBag Garden

Learning objective	Enable pupils to compare growbag plant position, growth and care using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Sack GrowBag Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils compare at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-175 - RECORD - Sack GrowBag Garden

Learning objective	Enable pupils to record growbag plant position, growth and care using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Sack GrowBag Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils record at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-176 - MAP - Sack GrowBag Garden

Learning objective	Enable pupils to map growbag plant position, growth and care using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Sack GrowBag Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils map at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-177 - GROW - Sack GrowBag Garden

Learning objective	Enable pupils to grow growbag plant position, growth and care using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Sack GrowBag Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils grow at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-178 - CHECK - Sack GrowBag Garden

Learning objective	Enable pupils to check growbag plant position, growth and care using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Sack GrowBag Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils check at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-179 - EXPLAIN - Sack GrowBag Garden

Learning objective	Enable pupils to explain growbag plant position, growth and care using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Sack GrowBag Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils explain at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-180 - IMPROVE - Sack GrowBag Garden

Learning objective	Enable pupils to improve growbag plant position, growth and care using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Sack GrowBag Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils improve at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 09 - Ground Production Garden

Teaching focus: rows, spacing, establishment and harvest.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Ground Production Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-181 - OBSERVE - Ground Production Garden

Learning objective	Enable pupils to observe rows, spacing, establishment and harvest using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Ground Production Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils observe at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-182 - COUNT - Ground Production Garden

Learning objective	Enable pupils to count rows, spacing, establishment and harvest using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Ground Production Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils count at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-183 - MEASURE - Ground Production Garden

Learning objective	Enable pupils to measure rows, spacing, establishment and harvest using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Ground Production Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils measure at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-184 - COMPARE - Ground Production Garden

Learning objective	Enable pupils to compare rows, spacing, establishment and harvest using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Ground Production Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils compare at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-185 - RECORD - Ground Production Garden

Learning objective	Enable pupils to record rows, spacing, establishment and harvest using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Ground Production Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils record at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-186 - MAP - Ground Production Garden

Learning objective	Enable pupils to map rows, spacing, establishment and harvest using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Ground Production Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils map at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-187 - GROW - Ground Production Garden

Learning objective	Enable pupils to grow rows, spacing, establishment and harvest using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Ground Production Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils grow at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-188 - CHECK - Ground Production Garden

Learning objective	Enable pupils to check rows, spacing, establishment and harvest using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Ground Production Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils check at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-189 - EXPLAIN - Ground Production Garden

Learning objective	Enable pupils to explain rows, spacing, establishment and harvest using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Ground Production Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils explain at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-190 - IMPROVE - Ground Production Garden

Learning objective	Enable pupils to improve rows, spacing, establishment and harvest using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Ground Production Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils improve at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 10 - Discovery Production Garden

Teaching focus: student-generated production questions.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Discovery Production Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-191 - OBSERVE - Discovery Production Garden

Learning objective	Enable pupils to observe student-generated production questions using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Discovery Production Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils observe at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-192 - COUNT - Discovery Production Garden

Learning objective	Enable pupils to count student-generated production questions using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Discovery Production Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils count at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-193 - MEASURE - Discovery Production Garden

Learning objective	Enable pupils to measure student-generated production questions using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Discovery Production Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils measure at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-194 - COMPARE - Discovery Production Garden

Learning objective	Enable pupils to compare student-generated production questions using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Discovery Production Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils compare at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-195 - RECORD - Discovery Production Garden

Learning objective	Enable pupils to record student-generated production questions using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Discovery Production Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils record at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-196 - MAP - Discovery Production Garden

Learning objective	Enable pupils to map student-generated production questions using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Discovery Production Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils map at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-197 - GROW - Discovery Production Garden

Learning objective	Enable pupils to grow student-generated production questions using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Discovery Production Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils grow at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-198 - CHECK - Discovery Production Garden

Learning objective	Enable pupils to check student-generated production questions using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Discovery Production Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils check at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-199 - EXPLAIN - Discovery Production Garden

Learning objective	Enable pupils to explain student-generated production questions using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Discovery Production Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils explain at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-200 - IMPROVE - Discovery Production Garden

Learning objective	Enable pupils to improve student-generated production questions using simple observations, counts or measurements and record what they find.
Materials	Notebook/journal, pencil, ruler where appropriate, tally sheet, labels, simple containers and teacher-approved garden tools.
DIY / low-cost option	Make rulers/markers from cardboard, use reused transparent containers for collection demonstrations, and create tally sheets on scrap card.
Teacher preparation	Mark two or more safe observation points in the Discovery Production Garden; check that tools are clean and measurements can be made without damaging plants.
Step-by-step delivery	1. Introduce the question. 2. Demonstrate the tool or observation method. 3. Pupils improve at assigned points. 4. Record at least two pieces of evidence. 5. Compare results. 6. Ask pupils to explain one pattern. 7. Complete a safe care action.
Questions to ask	What did you measure or count? Which result is larger/smaller? What might explain the difference? What should we check next?
Expected evidence	Tally, labeled drawing, measurement table, simple graph or short evidence-based explanation.
Differentiation	Use paired work, pre-drawn tables and fewer samples for pupils needing support; extension pupils add another sample or comparison.
Troubleshooting	If measurements differ widely, repeat the demonstration and check tool position, units and sample identity.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

LEVEL IMPLEMENTATION REVIEW

Use this page after completing the level's practical sequence.

Strongest learner competency	
Competency needing reteaching	
Garden/system needing maintenance	
Materials to replenish	
Safety or accessibility improvement	
Next mission/assessment priority	